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January 13, 1978
SUPERSEDING
Int. Fed. Spec. P-C-001121A
October 3, 1967

FEDERAL SPECIFICATION

CLEANING AND POLISHING COMPOUND, STAINLESS STEEL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a solvent-type cleaner and polish for stainless steel, which can also be used as an anti-stick or release type material.

1.2 Classification. The product shall be of the following types as specified (see 6.2):

Type I - Aerosol
Type II - Bulk (liquid)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking For Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific prccurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Hazardous Substances and Articles; Administration and Enforcement Regulations.

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(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standard:

E 168 - General Techniques of Infrared Quantitative Analysis.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The type I (liquid components) and the type II products shall consist of 1,1,1-trichloroethane and silicone oil. The percentage of 1,1,1-trichloroethane shall be 92 percent (minimum) to 98 percent (maximum). The percentage of silicone oil shall be 2 percent (minimum) to 8 percent (maximum). The quantitative analysis of the products shall be performed as specified in 4.4.1.3.

3.2 Salient features. The product shall have no discernable odor other than that of 1,1,1-trichloroethane. The product shall be completely colorless. The salient features of the product shall be examined as specified in 4.4.1.4.

3.3 Tackiness. When tested as specified in 4.4.1.5, the product shall not cause tackiness or gumminess on a test surface.

3.4 Label. The unit container label shall include the directions for use and the appropriate warning and caution statements, including a statement to the effect that the product is to be kept away from floors because of the slipperiness hazard. The label shall be in accordance with 16 CFR 1500.

3.5 Aerosol container (type I product).

3.5.1 Propellant. The use of fluorocarbons or vinyl chloride in the propellant gas is prohibited. The propellant shall be analyzed for fluorocarbons and vinyl chloride as specified in 4.4.1.3.2.

3.5.2 Valve performance. When tested as specified in 4.4.2.3, the aerosol valve shall deliver 0.75 g/second (minimum) to 1.35 g/second (maximum) at 27° ± 0.5°C.

3.5.3 Valve leakage. When tested as specified in 4.4.2.4, the valve leakage measured to the nearest ml shall be less than 0 ml in 48 hours.

3.5.4 Net weight. The measured net weight, determined as specified in 4.4.2.5 shall not be less than the net weight specified on the container label.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Inspection sampling. Sampling for the inspection procedures set forth in this specification shall be performed in accordance with MIL-STD-105. The lots to be inspected shall consist of no more than one classification (see 1.2) and one package size (see 5.1).

4.3 Examination.

4.3.1 Examination of preparation for delivery.

4.3.1.1 Sampling plan. The sample unit shall be defined as one shipping container fully marked and prepared for delivery. The samples shall be selected in accordance with inspection level S-2, and the AQL shall be 4.0 percent defective.

4.3.1.2 Examination procedure. An examination shall be made to determine the compliance of the packing and marking with the requirements of 5.1 and 5.3. The sample units shall be examined for the defects described in table I and scored accordingly.

TABLE I. Examination of preparation for delivery

Examine	Defects
Container	Not as specified.
Content	Not as specified; incorrect number of packages within.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.2 Examination of the package.

4.3.2.1 Sampling plan. The sample unit shall be defined as one such package including its label. The samples shall be selected in accordance with inspection level S-2, and the AQL shall be 4.0 percent defective.

4.3.2.2 Examination procedure. The examination shall be made to determine the compliance of the packaging and marking with the requirements of 5.2 and 5.3. The sample units shall be examined for the defects described in table II and scored accordingly.

TABLE II. Examination of the package

Examine	Defects
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.4 Testing.

4.4.1 Product testing.

4.4.1.1 Sampling plan.

4.4.1.1.1 Type I. One container shall be used for the chemical analysis (see 4.4.1.3) and salient features (see 4.4.1.4), and the other container shall be used in the tackiness test (see 4.4.1.5). The lot shall be unacceptable if the product fails any of the tests described in 4.4.1.

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4.4.1.1.2 Type II. A 1-quart sample shall be used. The lot shall be unacceptable if the composite fails any of the applicable tests described in 4.4.1.

4.4.1.2 Testing procedures. The tests described below shall be made to determine compliance with the chemical requirements of 3.1.

4.4.1.3 Chemical analysis. All weights taken in this procedure shall be to the nearest 0.01 gram.

4.4.1.3.1 Special apparatus and materials. The chemical analysis procedure described herein requires the following apparatus and materials:

A can piercing, pressure measuring device recommended in the CSMA Aerosol Guide for determining the internal pressure of aerosol products. The device is available as Model CBA-6 from Builders Sheet Metal Works, Inc., 110 Wooster Street, New York, New York 10012.

A temperature programmable gas chromatograph (Hewlett-Packard Model 5750 gas chromatograph, or equivalent) equipped with a 0.5 ml gas sampling valve and thermal conductivity detector.

Two stainless steel gas chromatography columns, 6 feet long by 1/8" outside diameter, packed with 60-200 mesh silica gel.

Pure propellant gases in lecture bottles from Matheson Gas Products (or another supplier).

Matheson lecture bottle control valve Model 58, or equivalent.

4.4.1.3.2 Gas chromatographic analysis. Condition the chromatographic columns at 200°C with helium as the carrier gas at 75-100 ml/minute for one hour. Remove the actuator and cap of an unused aerosol container and weigh the container. Puncture the can using the pressure measuring device and connect the device to the gas chromatograph. Bubble the gas sample through the sampling valve for two minutes before each run is made to purge the valve of any other gases. Analyze the propellant using a thermal/conductivity detector. The temperature program shall be 100°C to 200°C at 6°C per minute with a 2-minute post injection hold at 100°C and a 5 minute upper limit hold at 200°C. Helium is used as the carrier gas at the rate 75-100 ml per minute.

Release the remainder of the propellant through the pressure measuring device. Open the container and pour at least 50 gm of the liquid into a bottle, seal against evaporation, and weigh the bottle and contents.

Determine the composition of the propellant using the gas chromatogram of the product and chromatograms of the standard propellants obtained using the gas chromatographic analysis procedure as described above. There shall be no detectable material in the aerosol propellant sample having the same retention time as: vinyl chloride, fluorocarbon Nos. 11, 12, 114, 115.

4.4.1.3.3 Quantitative analysis. From the bottle containing the liquid product, pour approximately 30 gm into a weighed evaporating dish. Quickly recap the bottle and weigh to the nearest 0.01 gm to calculate the amount poured into the evaporating dish. Place the evaporating dish into a circulating air oven at 85° ± 0.5°C for 2-1/2 hours. Remove the dish, cool in a desiccator, and weigh. The difference between this weight and the weight of the empty dish is the weight of the residue. Calculate the percentage of residue in the liquid sample. The percentage of 1,1,1-trichloroethane shall be reported as the difference between 100 and the percentage of residue in the product. Save the residue in a sealed container for infrared analysis (see 4.4.1.3.4).

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4.4.1.3.4 Infrared analysis.

4.4.1.3.4.1 Analysis of liquid sample. Take approximately 15 ul from the bottle of liquid sample (see 4.4.1.3.2) and place it in a 10 cm gas cell with KBr windows. Immediately close the cell and allow the contents to reach equilibrium (about 5 minutes) inside the sample compartment of the spectrophotometer. Obtain the spectrum of the sample from 4000 cm-1 to 625 cm-1. Measure the absorbances of the peaks using the baseline method as explained in ASTM E 168. Determine the proportioning factor, P, using the following formula:

$$P = \frac{ASM}{AST}, \text{ where}$$

ASM = Absorbance of the corresponding peak in the sample spectrum.
AST = Absorbance of one of the larger peaks in the standard spectrum of trichloroethane (see figure 1).

For each of the propellant components, as determined in 4.4.1.3.2, obtain an infrared spectrum and determine a proportioning factor, using the above formula, to account for propellant peak absorbances in the liquid sample spectrum. For acceptance, absorbances of the standard spectrum and pure propellant spectra, when multiplied by their appropriate proportioning factors, shall match all of the absorbances in the liquid sample spectrum within 2 percent. Any other peaks in the liquid sample spectrum not accounted for by the standard spectra shall constitute failure. Disregard typical carbon dioxide and water vapor peaks, if present.

4.4.1.3.4.2 Analysis of the residue. Press a small amount of the residue from the quantitative analysis procedure (see 4.4.1.3.3) between two polished KBr flats. Obtain the spectrum of the sample from 4000 cm-1 to 625 cm-1. Measure the absorbances of the peaks using the baseline method as explained in ASTM E 168. Determine the proportioning factor, P, using the following formula:

$$P = \frac{ASM}{AST}, \text{ where}$$

ASM = Absorbance of one of the larger peaks in the sample spectrum (residue).
AST = Absorbance of the corresponding peak in the standard spectrum of silicone oil (see figure II).

For acceptance, absorbances of the standard spectrum, when multiplied by P, shall match all of the absorbances in the residue spectrum within 2 percent. Any other peaks in the residue spectrum not corresponding within 2 percent shall constitute failure. Disregard typical carbon dioxide and water vapor peaks, if present.

4.4.1.4 Salient features. The bottle of liquid sample from the chemical analysis procedure (see 4.4.1.3) shall be examined for the presence of any color. The odor of the liquid shall also be compared to that of a container of 1,1,1-trichloroethane.

4.4.1.5 Tackiness. Spray the product on a clean glass plate for 1 second (or 1 gm of type II) and allow to remain in air at room temperature for 24 hours. Examine the surface by touch. The surface may be slightly oily but shall not have any evidence of gumminess or tackiness.

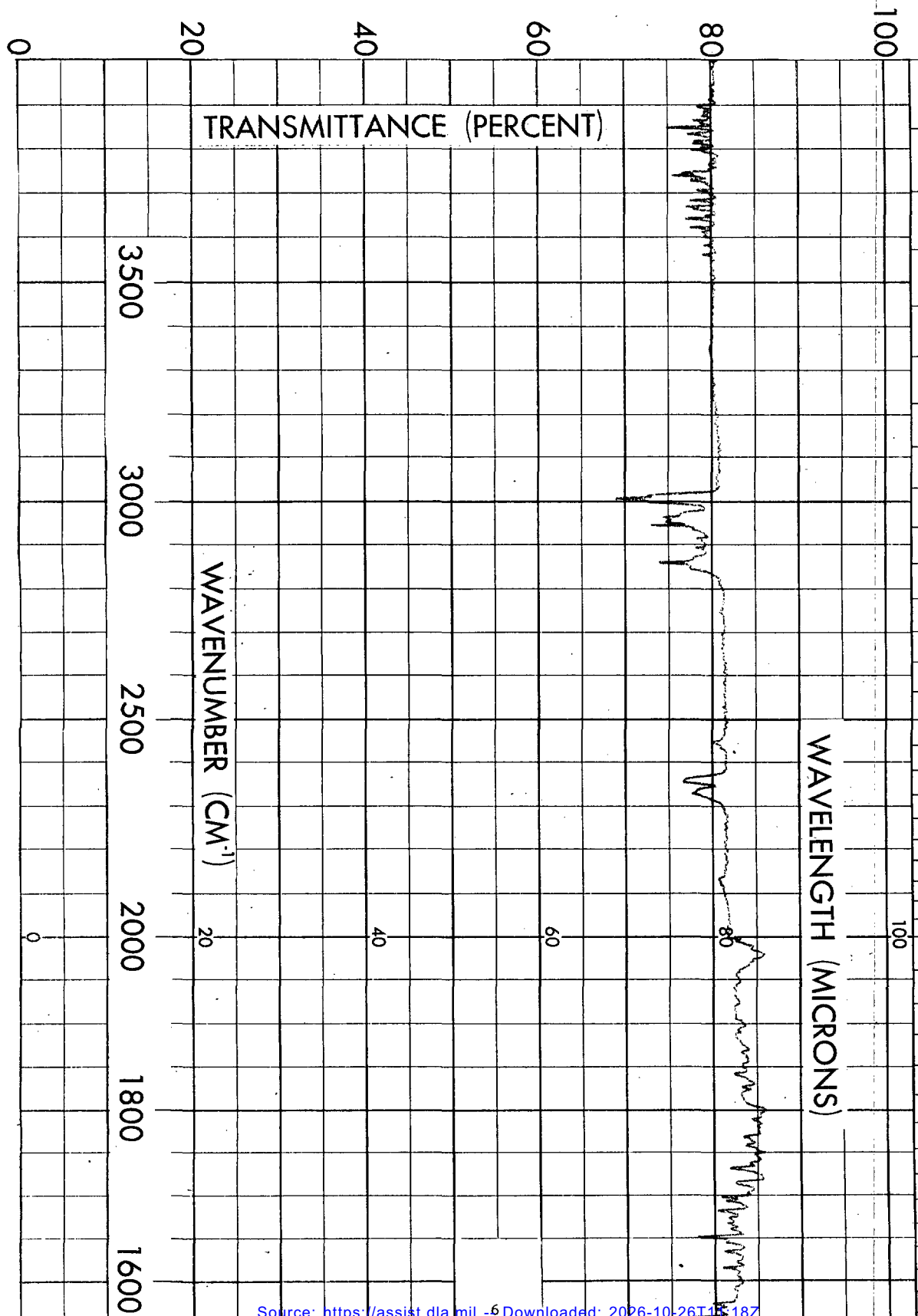
4.4.2 Aerosol container testing.

4.4.2.1 Sampling plan. The sample unit shall be defined as three aerosol containers. The sample units shall be taken from the lot in accordance with inspection level S-1. From each sample unit, one container shall be used in each of the three tests described in 4.4.2. The AQL shall be 4.0 percent defective.

4.4.2.2 Testing procedures. The tests described below shall be made to determine compliance with paragraph 3.6.

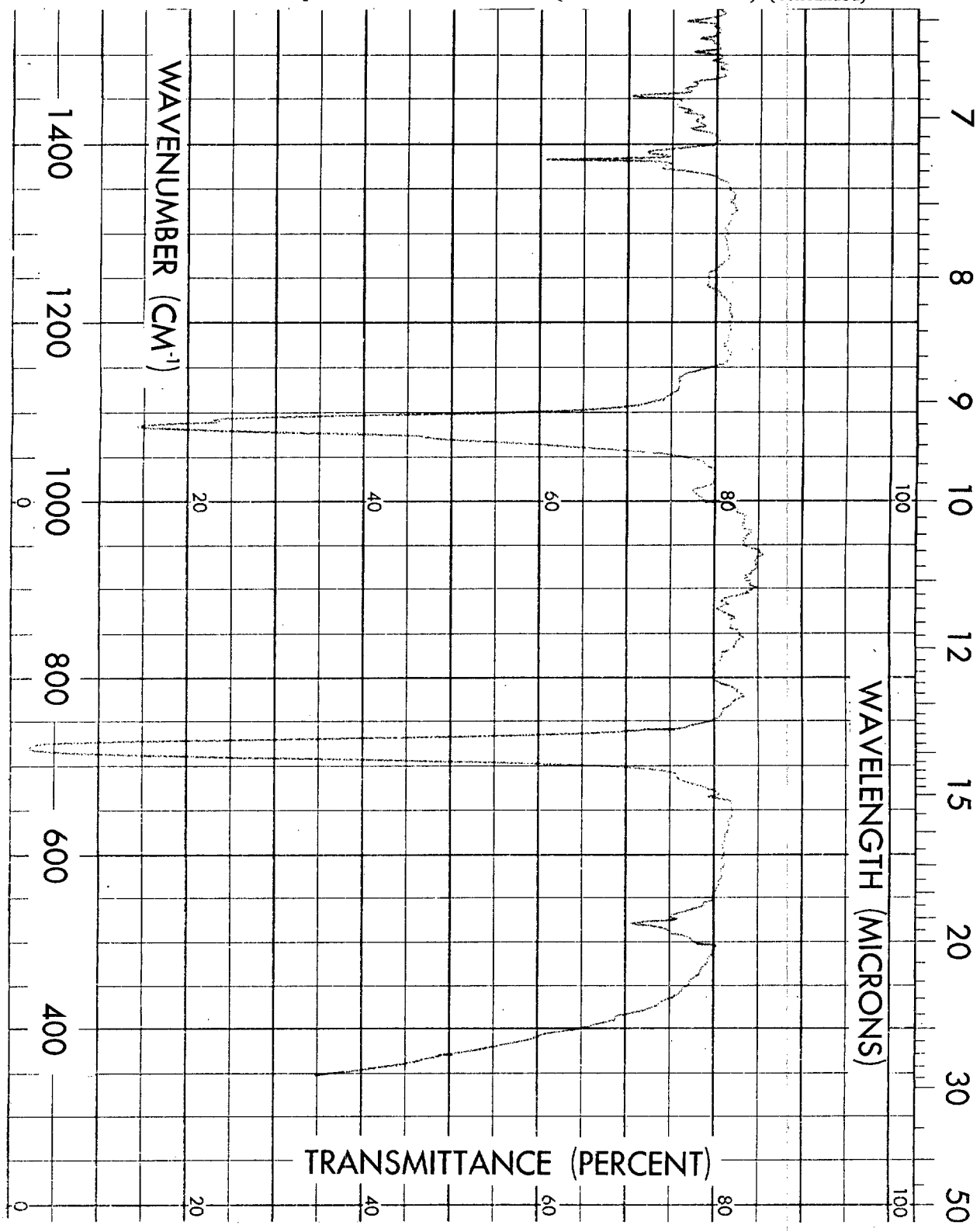
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Figure I. Standard Spectrum of Trichloroethane (with 5% Silicone oil)



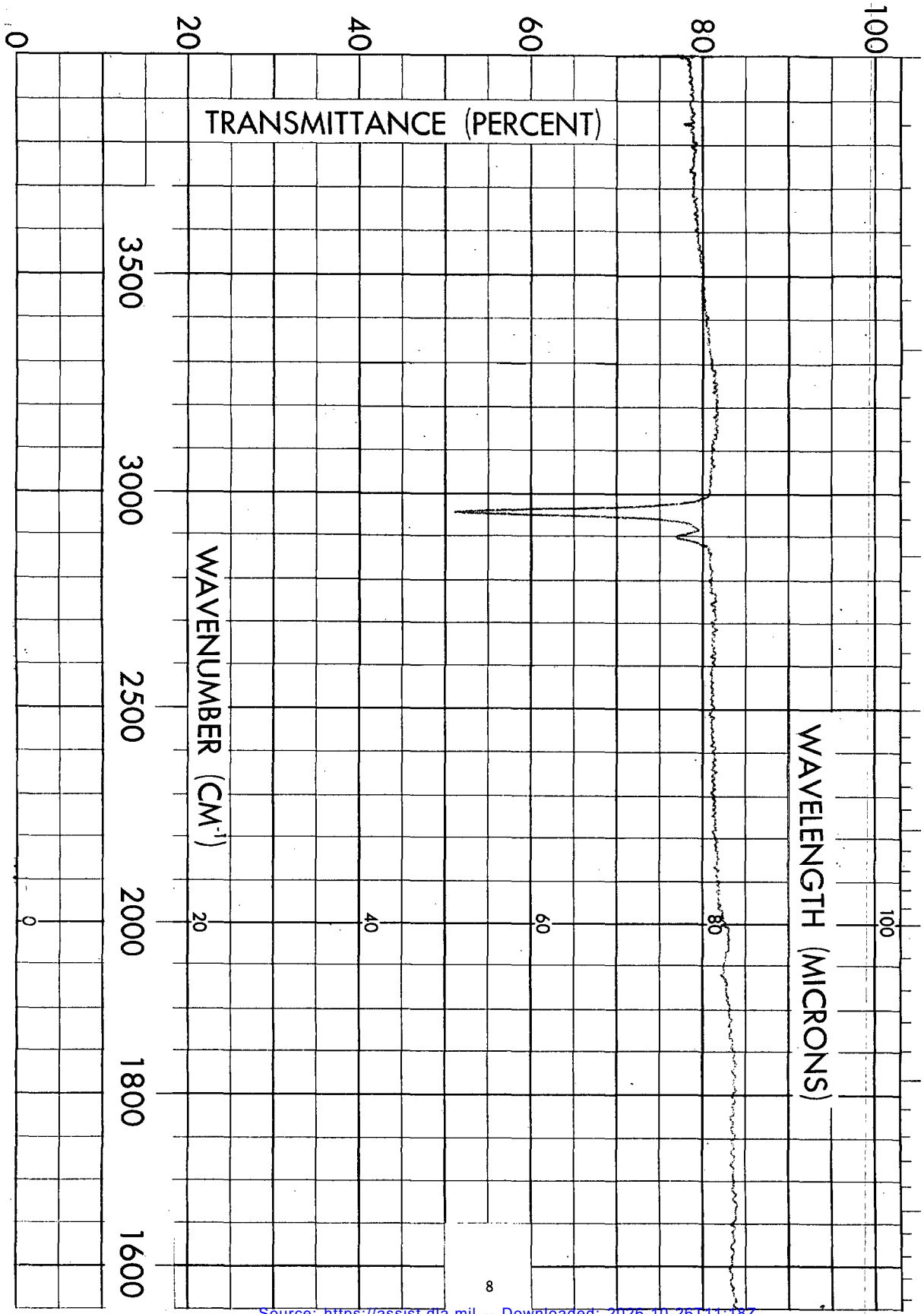
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Figure I. Standard Spectrum of Trichloroethane (with 5% Silicone oil) (continued)



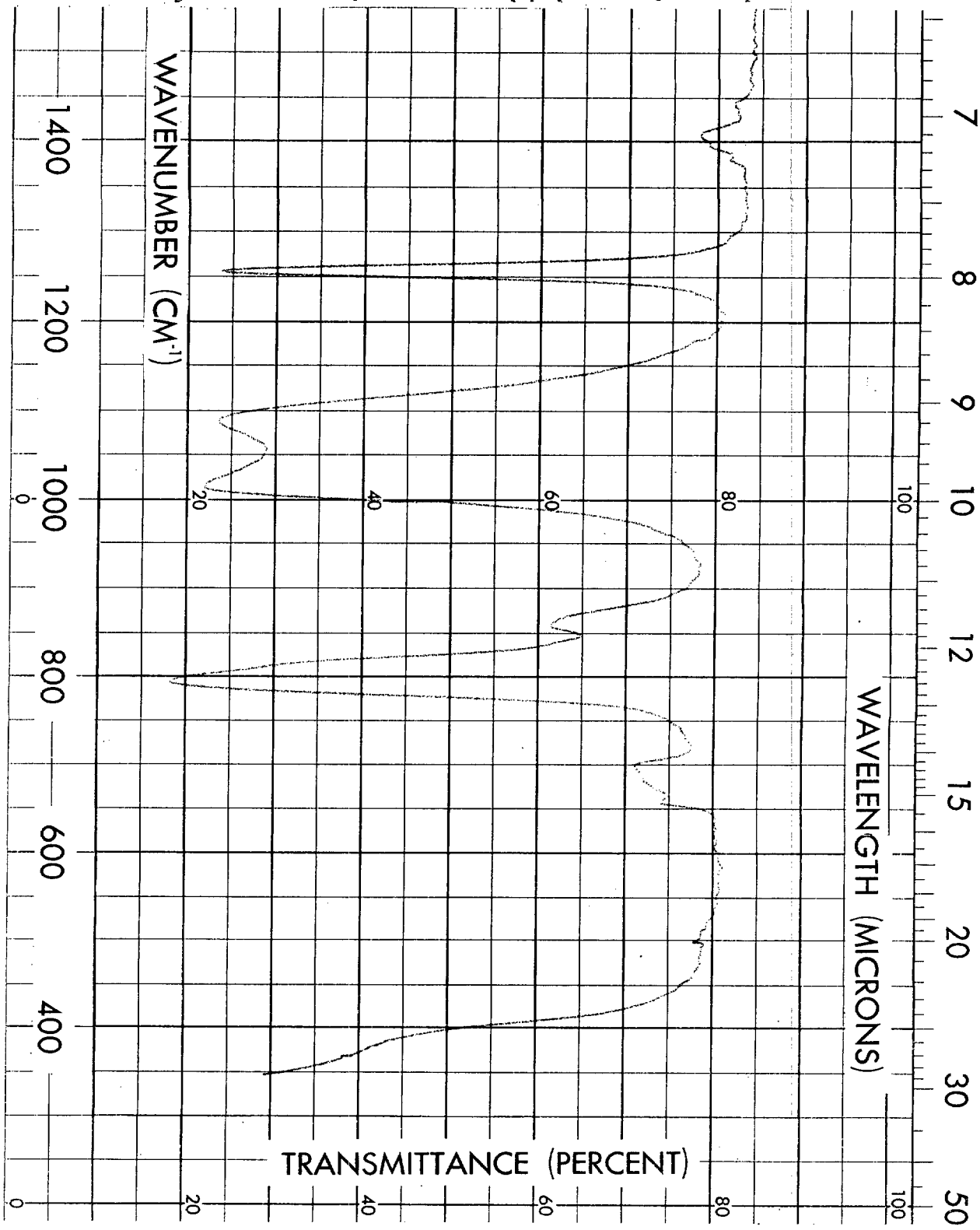
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Figure II. Standard Spectrum of Dimethylpolysiloxane (viscosity at 25°C is 350 cp)



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Figure II. Standard Spectrum of Dimethylpolysiloxane (continued)



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4.4.2.3 Valve performance. Remove paper label, protective cover, and all other detachable materials except for the actuator. Activate the valve for a few seconds and weigh the container to the nearest 0.05 g. Place the container in a water bath equipped with a stirrer and maintained at 20° ± 0.5°C (68° ± 1°F). The container shall be kept completely submerged for 30 minutes. Remove and quickly dry the dispenser with a towel and use a blast of compressed air to remove moisture from areas inaccessable to the towel. Activate the valve for 10.0 seconds. Reweigh the dispenser and compute the weight loss. The delivery rate shall be calculated as the weight loss in grams, ± 0.1 gram, divided by the test spray time (10.0 seconds) to 0.1 second accuracy. The delivery rate shall be reported as the weight, ± 0.1 g, delivered per second.

4.4.2.4 Valve leakage. For valve leakage, a 1 ounce vial calibrated in milliters and completely filled with water shall be placed over the valve of an unused aerosol container for 48 hours. During this time, the aerosol container and vial shall be placed in water to just above the lower edge of the vial. The water level in the vial shall be recorded at the beginning and end of the test. Any lowering of the water level during the test shall constitute failure.

4.4.2.5 Net weight. Weigh filled container to the nearest gram. Shake the container with a wrist twisting motion for 15 seconds at the approximate rate of one complete cycle per second. Holding the container in the position recommended on the product label, exhaust the container by holding the valve wide open. Rinse the exterior of the container and dry, using a towel and compressed air for drying inaccessable areas including the interior of the actuator. Weigh the exhausted container to the nearest gram. The measured net weight in ounces shall be calculated using the following formula:

$$N.W. = (G - E - 10)/28.35$$

Where: G = gross weight of filled container (grams).
E = weight of exhausted container (grams).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Unless otherwise specified, the product shall be preserved and packaged in accordance with normal commercial practice. The type I product shall be packaged in aerosol containers which shall conform to the requirements of 3.5. The quantity of product supplied per packaged shall be as specified in the procurement document (see 6.2). The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Unless otherwise specified, the packaged compound shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance with 3.4, Hazardous Substances Regulations, and Federal Standard No. 123, as applicable (see 6.2).

6. NOTES

6.1 Intended use. This aerosol provides a nonflammable cleaner and polish for stainless steel. The solvent base provides degreasing properties. In addition, the product containing silicones provides anti-stick or release properties which act as a stable lubricant.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Packaging and packing required (see 5.1 and 5.2).
(d) Marking required (see 5.4).

MILITARY INTEREST:

The Department of Defense has waived coordination interest in this document.

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITY:

CPSC

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